

Q1: Multiple choice questions

(2pts)

1. "Software as a Service (SaaS)" slides list one benefit as pay-as-you-go and accessibility, and one drawback as:

☐ Guaranteed performance

☐ Full developer control

☒ Security/trust vulnerabilities and internet dependence

☐ Free on-prem hardware

Software as a Service (SaaS)

- Benefits
 - Pay as you go (monthly, annually)
 - Accessible anywhere
 - No need for expensive computers
- Drawbacks
 - Lack of control
 - Vulnerable security and trust
 - Limited range of applications
 - Internet dependent
 - Performance issues

2. According to "Application types," what determines which engineering methods/tools to use?

☐ Project manager's personal preference

☒ Application type, customer requirements, team background, and tools

☐ Software cost alone

☐ All of the above

Application types

Software engineering diversity

- There is a wide diversity of types of software engineering projects, from development of software systems for a customer to software engineering.
- The software engineering methods and tools used for software engineering are determined by the type of software engineering project.
- The key to software engineering diversity is the requirement of the customer.
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3. A key weakness of the Waterfall model in the slides is:

☐ Low documentation

☐ Iterative

☐ Lack of any structure

☒ Inflexible to changing requirements and late problem discovery

The waterfall model

- Strengths
 - Aligns to system engineering process
 - Complete set of documentation
- Weaknesses
 - Inflexible to changing requirements
 - Late discovery of technical problems
 - Sequential model phase works for previous phase

4. To cope with change, there are two approaches

☐ Code-and-fix & Big-bang integration

☐ Change prevention & big-bang integration

☒ Prototyping & incremental delivery

☐ Waterfall with extended design phase & late verification

Coping with change (cont'd)

- Two approaches are used to cope with requirements change:
 - Incremental delivery** is used to cope with change by producing partially required services for early feedback.
 - Waterfall model** is used to cope with change by producing a complete set of services and then making changes to the system.

Q2: For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR). (4pts)

Requirement	User vs. System	FR vs. NFR
"The system should be <u>easy to use</u> for first-year students."	User	NFR
"The application shall <u>allow</u> password reset using a verified email."	System	FR
"The report generation shall be <u>fast</u> ."		NFR
"The interface will <u>display</u> student GPA to two decimal places."	System	FR

Q3: Team of 6 people; workday = 8h. Last 4 sprints completed story points: 22, 28, 26, 24.

1. Compute the sprint velocity (use the average sprint story points).

$$\text{Sprint Velocity} = (22+28+26+24)/4 = 25$$

2. Compute focus factor ($\text{Sprint Velocity} \div (\text{people} \times \text{hours/day})$). (1pts)

$$\text{Focus Factor} = 25 / 48 = 0.52 \text{ (not optimal range)}$$

because optimal range is 0.6 to 0.8

3. For a 3-week sprint (Mon–Fri, weekends off), compute effective team capacity (hours) using the focus factor. Show formulas and round sensibly. (1pts)

$$3 \text{ week} = 5+5+5 = 15 \text{ (احذفوا يومين من كل اسبوع)}$$

$$(48 * 0.52 * 15) = 375$$

4. If the next sprint backlog is 30 story points, state whether the plan is feasible and why.

$$30 > 25 \text{ (prudent)}$$

Feasibility: ☐ Feasible

or

☒ Prudent

(1pts)

Justification:

Backlog $\approx$ Velocity $\rightarrow$ Feasible

Backlog $>$ Velocity (يفرق بسيط) $\rightarrow$ Prudent

Backlog $>>$ Velocity $\rightarrow$ Not Feasible

Q4: For each project, choose one development process model: Agile, Plan-Driven, or Component-Based. (4pts)

Project 1 — Smart-City Sensor Hub

- Stakeholder Presence: Moderate; city IT holds bi-weekly reviews; end-users (operators) rarely available.
- Requirement Volatility: Medium; new sensor types/vendors appear mid-project, but core data collection goals are stable.
- Project Size & Team: 20 engineers across 3 vendors; distributed.
- Architecture Constraints: Must integrate COTS gateways and vendor SDKs; strict uptime (99.9%).
- Compliance: Must pass basic cybersecurity hardening but no heavy certification.

Choose: ☐ Agile

☒ Plan-Driven

☒ Component-Based

Project 2 — Banking Regulatory Reporting

- Stakeholder Presence: Limited; auditors and compliance office set requirements up front and review only at stage gates.
- Requirement Volatility: Low; templates and KPIs fixed by regulation for the next 3 years.
- Project Size & Team: 30 people, multi-department; formal PMO.
- Documentation & Traceability: Detailed SRS/SDD, audit trails, sign-offs required.
- Risk Profile: High penalty for defects; schedule predictable.

Choose: ☐ Agile

☒ Plan-Driven

☒ Component-Based

Project 3 — Campus Social-Learning MVP

- Stakeholder Presence: High; product owner and pilot students give daily feedback.
- Requirement Volatility: High; features frequently reprioritized to find product-market fit.
- Project Size & Team: 6 developers, co-located; CI/CD in place.
- Time to Market: Need public beta in 4-6 weeks; success measured by adoption/engagement.
- Architecture: Greenfield; minimal legacy constraints.

Choose: ☒ Agile

☐ Plan-Driven

☐ Component-Based

Project 4 — Medical Device Companion App

- Stakeholder Presence: Moderate; clinicians available for periodic usability reviews; regulatory team sets constraints.
- Requirement Volatility: Medium; UI iterations expected, but clinical workflows and data formats are fixed by standards.
- Project Size & Team: 15 people; UX + mobile + back-end.
- Compliance: Must align with IEC 62304 software life-cycle documentation and ISO 14971 risk management; app is Class II accessory.
- Integration: Uses a certified SDK from the device vendor for data acquisition.

Choose: ☐ Agile ☒ Plan-Driven ☐ Component-Based

Q5: For each statement, indicate the requirement type: Product, Organization or External (6pts)

1. The API shall respond in ≤ 300 ms (p95) for read requests under 100 concurrent users.

Choose: ☒ Product ☐ Organization ☐ External

2. All new services must be built in Java 21 and deployed on the organization's Kubernetes platform.

Choose: ☐ Product ☒ Organization ☐ External

3. Patient data must comply with HIPAA and local privacy law X.

Choose: ☐ Product ☐ Organization ☒ External

4. The mobile app shall support offline read-only access for 24 hours.

Choose: ☒ Product ☐ Organization ☐ External

5. The system shall export reports in CSV and PDF.

Choose: ☒ Product ☐ Organization ☐ External

6. Device firmware must interoperate with ISO/IEC 14443 Type A NFC readers.

Choose: ☒ Product ☐ Organization ☐ External